RAG Learning

2.5 Statistical Hypothesis Testing

Mathematics B (MEI) — OCR B — A-Level (H640) — Statistics — spec refs H1–H11

What this topic covers. The language and logic of hypothesis testing, tests on a binomial probability, the distribution of the sample mean and tests on a population mean using the Normal distribution, and using a given correlation coefficient to make an inference.

1. The language

Term	Meaning
Null hypothesis H_0	The default assumption being tested — always an equality.
Alternative hypothesis H_1	What is concluded if H_0 is rejected.
Test statistic	The quantity calculated from the sample and compared against the critical value.
Significance level	The probability of incorrectly rejecting H_0.
Critical region	The values of the test statistic that lead to rejection. Also called the rejection region.
Acceptance region	The values for which H_0 is not rejected.
Critical value	The boundary of the critical region.
p-value	The probability of a result at least as extreme as the one observed, assuming H_0 is true.

Hypotheses are always statements about a **population parameter** — p or μ — never about the sample. Writing $H_0 : \bar{x} = 50$ is wrong; it must be $H_0 : \mu = 50$.

2. One-tailed or two-tailed?

Wording	Test	H_1
"biased towards heads", "has increased"	One-tailed	$p > 0.5$
"has decreased", "is less effective"	One-tailed	$p < 0.5$
"is biased", "has changed"	Two-tailed	$p \neq 0.5$

For a two-tailed test at the α level, the probability is split between the tails — so a 5% two-tailed test uses 2.5% in each tail.

The word that decides this is whether a **direction** is specified. "Biased" alone gives no direction and demands a two-tailed test; "biased towards heads" gives one.

3. The logic of the test

1. State H_0 and H_1 in terms of a population parameter.
2. Assume H_0 is true.
3. Calculate how likely the observed result — or something more extreme — would be under that assumption.
4. If that probability is below the significance level, the result is unlikely under H_0, so **reject** H_0.
5. If not, there is **insufficient evidence** to reject H_0.
6. State the conclusion in the **context** of the original question.

Never write "accept H_0". Failing to reject is not proof of truth — it means the evidence was not strong enough. Similarly, rejecting H_0 gives **evidence**, never proof. Both phrasings are penalised.

The significance level is the probability of **incorrectly rejecting** a true null hypothesis (spec ref H3). A 5% level means that, if H_0 is true, there is at most a 5% chance of wrongly rejecting it.

4. Binomial hypothesis tests

Here H_0 takes the form $p = \text{a particular value}$, and the test statistic is the observed number of successes (H4–H6).

Worked example. A coin is suspected of being biased towards heads. Set up the test.

$$H_0 : p = 0.5, \qquad H_1 : p > 0.5$$

One-tailed, since a direction is specified. Reject H_0 if the probability of getting at least the observed number of heads is below the significance level.

Worked example. A test at the 5% level gives a p-value of 0.032. State the conclusion.
Since $0.032 < 0.05$, reject H_0. There is evidence at the 5% level to support the alternative hypothesis, stated in context.

Worked example. The same test gives a p-value of 0.08.
Since $0.08 > 0.05$, do not reject H_0. There is insufficient evidence at the 5% level.

Because the binomial distribution is **discrete**, the probability of landing in the rejection region is usually *less* than the intended significance level, and only rarely equal to it. MEI notes this explicitly and states that learners are not tested on the distinction — but it explains why critical regions look slightly "off".

5. The distribution of the sample mean

For random samples of size n drawn from $\mathrm{N}(\mu, \sigma^2)$, the sample mean is itself Normally distributed (H7):

$$\bar{X} \sim \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

so its standard deviation is $\dfrac{\sigma}{\sqrt{n}}$.

For $X \sim \mathrm{N}(100, 36)$ and samples of size 9, the sample mean has variance $\frac{36}{9} = 4$ and standard deviation 2.

The **Central Limit Theorem** extends this: for a large sample, the sample mean is approximately Normal *even when the population is not*. This is what allows a Normal test on a mean in many practical situations.

Notice that increasing n shrinks $\dfrac{\sigma}{\sqrt{n}}$. A larger sample makes the sample mean cluster more tightly around μ, so genuine differences become easier to detect.

6. Testing a population mean

Valid when the population variance is known, or is unknown but the sample is large (H8). The significance level will be given.

$$z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Worked example. A sample of 25 from $\mathrm{N}(\mu, 100)$ has mean 44. Test $H_0 : \mu = 40$ against $H_1 : \mu > 40$ at the 5% level.
Under H_0, $\bar{X} \sim \mathrm{N}\left(40, \frac{100}{25}\right) = \mathrm{N}(40, 4)$, so the standard deviation is 2.

$$z = \frac{44 - 40}{2} = 2$$

The one-tailed 5% critical value is 1.645. Since $2 > 1.645$, the result lies in the critical region: reject H_0. There is evidence at the 5% level that the population mean exceeds 40.

A Normal test on a mean is **not** valid with a small sample and an unknown population variance — with $n = 8$ and σ unknown, neither condition is met.

7. Correlation

Correlation measures how close data points lie to a straight line; a **rank** correlation coefficient measures it between the ranks rather than the raw values (H10).

For a correlation test, the null hypothesis is that there is **no correlation** (or no association) in the population. You compare a *given* coefficient against a given critical value or p-value and draw the inference (H11).

What is excluded. Calculating a correlation coefficient is not required, and neither is knowing the names of particular coefficients. Questions supply the coefficient and indicate whether correlation or association is being tested. "Association" refers to a more general relationship than a straight-line one.

A significant correlation still does not establish **causation**. A third variable may drive both — the standard example being hot weather driving both ice cream sales and drownings.

8. Interpreting results honestly

- The significance level must be chosen **before** the data are examined.
- Lowering the level from 5% to 1% demands stronger evidence and makes rejection harder.
- A result significant at 5% but not at 1% has a p-value between 0.01 and 0.05.
- Running many tests at the 5% level on data with no real effect will still produce false positives — about 1 in every 20.

9. Common pitfalls

- Writing hypotheses about the sample statistic rather than the population parameter.
- Using a one-tailed test when no direction is specified.
- Forgetting to halve the significance level for each tail in a two-tailed test.
- Writing "accept H_0" or claiming something is "proved".
- Dividing by σ instead of $\frac{\sigma}{\sqrt{n}}$ when testing a mean.
- Omitting the conclusion in context.
- Using a Normal test with a small sample and unknown variance.
- Choosing the significance level after seeing the result.